

Phase III Notes: 3-Ball Cilia Exploration

May 30, 2026

1 Goal of this phase

The goal of this phase is to move beyond the two-ball benchmark and begin exploring a richer cilium model with three segments / three balls / three relative angles. The main motivation is that the two-ball benchmark, while very useful, seemed to support only a fairly narrow family of strong strokes. Adding one more joint broadens the accessible stroke space and gives a better test of which conclusions from the benchmark survive in a more expressive geometry.

I also want to begin in a regime where value iteration and tabular methods are still feasible, so that I can compare exact, tabular, and neural methods on the same benchmark.

2 3-ball environment definition

The current three-ball environment is intended as the simplest natural extension of the two-ball benchmark.

Geometry and state representation

The cilium is modeled as three equal-length segments with fixed total length:

$$L_1 = L_2 = L_3 = \frac{1}{3}.$$

As in the two-ball case, the environment stores *relative* angles,

$$\phi = (\phi_1, \phi_2, \phi_3),$$

where ϕ_1 is the base angle and the physical segment angles are reconstructed cumulatively:

$$\psi_1 = \phi_1, \quad \psi_2 = \phi_1 + \phi_2, \quad \psi_3 = \phi_1 + \phi_2 + \phi_3.$$

This preserves the same serial-link convention used in the two-ball code and keeps the interpretation of the coordinates consistent across phases.

Angle ranges and discretization

The current working discretization is

$$n_{\text{bins}} = [11, 21, 21],$$

with angle ranges

$$\phi_1 \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right], \quad \phi_2, \phi_3 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right].$$

This keeps the base-angle range comparable to the two-ball benchmark while allowing the internal joints enough freedom to produce richer motions.

Actions

The action space is the natural extension of the two-ball setting: each relative angle can increment by -1 , 0 , or 1 bin per step, excluding the all-zero action. Thus the total number of actions is

$$3^3 - 1 = 26.$$

Boundary handling and reward

The reward is still the same one-step flux-like quantity derived from the Stokes/image-system solve, together with the same general boundary-handling logic (`clip_penalty` or `stay_penalty`).

3 What turned out to matter immediately

Numerical safeguards

In the first three-ball experiments, the raw reward table contained very large outliers coming from near-singular hydrodynamic solves. Adding simple numerical safeguards—for example rejecting near-contact configurations and very large finite force/flux values—was necessary before value iteration became interpretable.

Physical wall admissibility

Even after the numerical blow-up was controlled, value iteration initially found a short local cycle that dipped below the wall. This made it clear that angle-box constraints alone are not enough once the geometry is richer: the environment also needs a physical admissibility condition.

The fix used here was to require that all ball centers remain at least one radius above the wall:

$$z_i > a$$

for each ball center, where a is the ball radius.

After adding this condition, the short degenerate cycle disappeared and value iteration produced a longer coordinated recurrent motion.

4 Value iteration on the three-ball benchmark

With the wall-clearance condition and the numerical safeguards in place, value iteration on the refined three-ball benchmark

$$[11, 21, 21]$$

produced a cycle of length 23 with average reward

$$0.797059.$$

This is the highest average reward recorded so far on the refined three-ball benchmark among the methods tested here. The corresponding cycle serves as the current benchmark reference.

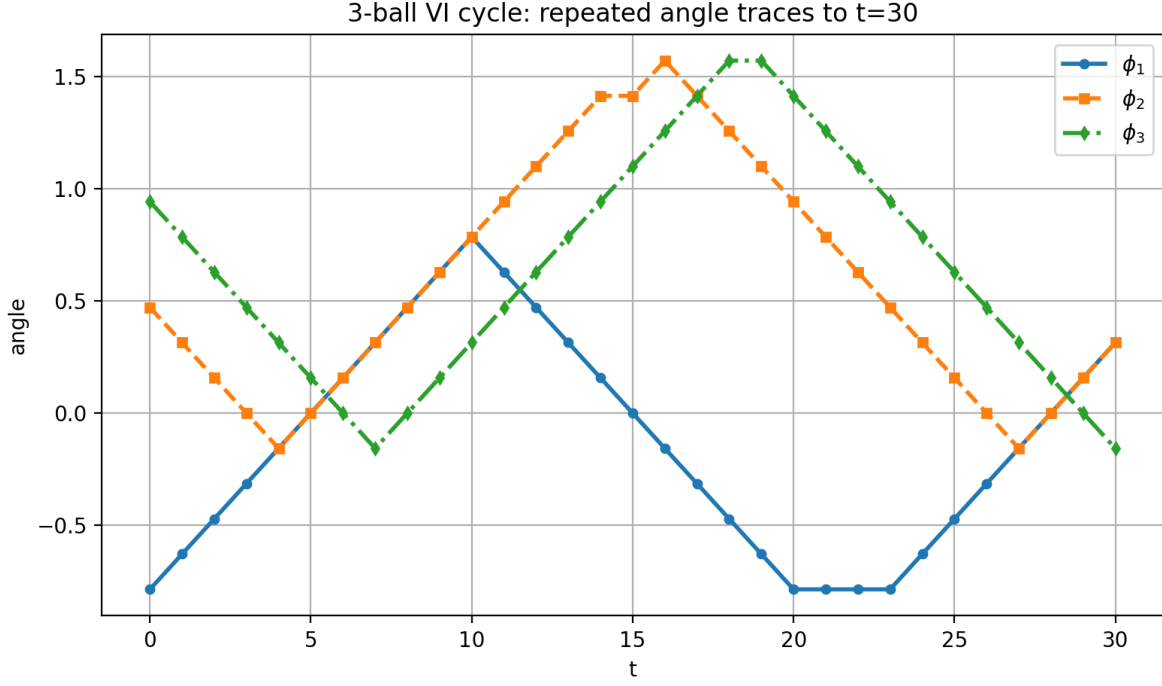


Figure 1: Repeated angle traces for the refined three-ball VI cycle.

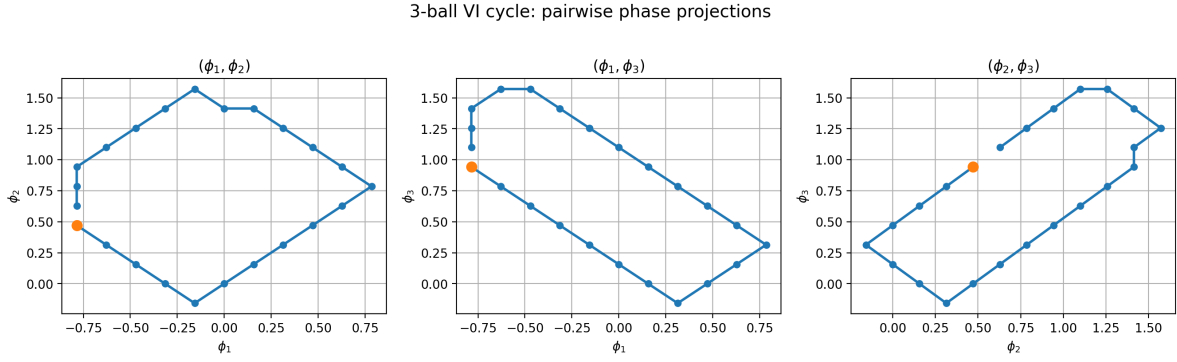


Figure 2: Pairwise phase projections for the refined three-ball VI cycle.

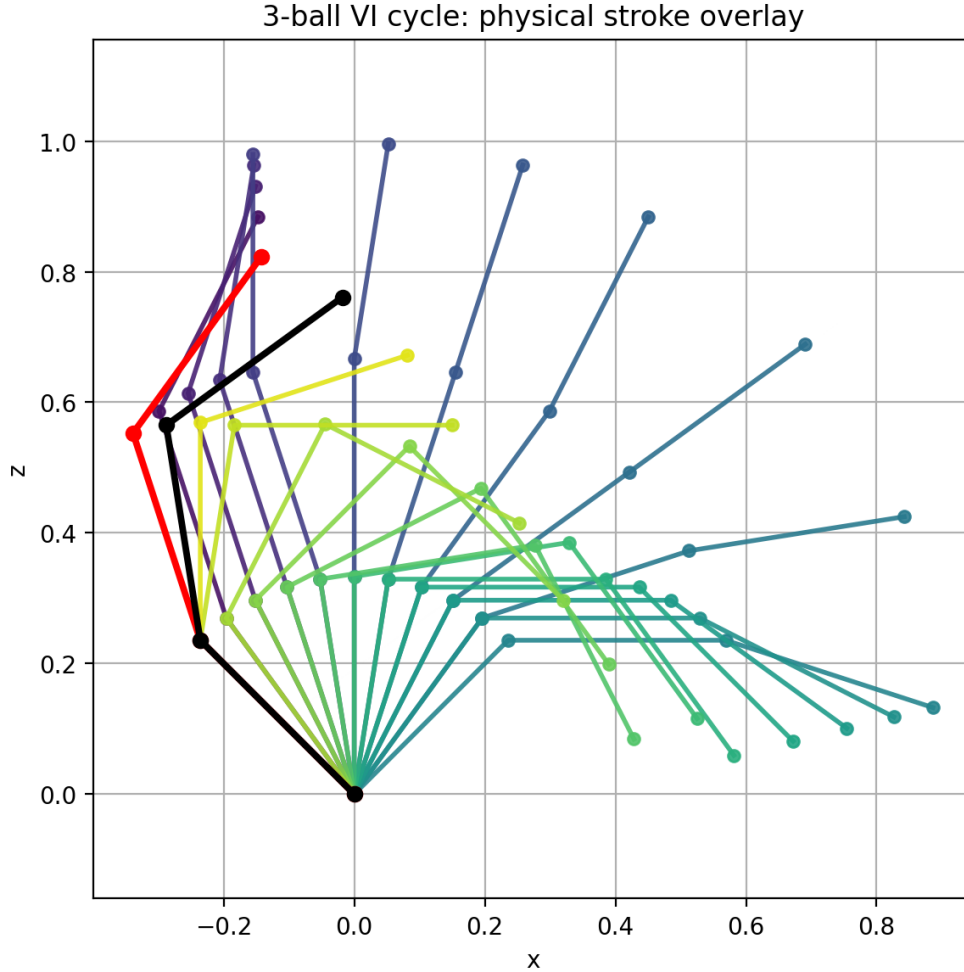


Figure 3: Physical stroke overlay for the refined three-ball VI cycle.

5 Tabular Q-learning on the refined three-ball benchmark

I next tested tabular Q-learning on the same refined three-ball environment with discretization

$$[11, 21, 21].$$

A practical point here was that the episode horizon matters. Shorter episode lengths often led to shorter or less satisfactory learned cycles, while longer runs with sufficiently long episodes produced a more coherent picture.

Across five seeds, the best tabular-Q cycles had lengths between 21 and 23, and average rewards between about 0.70 and 0.76. Thus the tabular-Q runs were close to the value-iteration benchmark in cycle length, but did not exceed the VI average reward of 0.797059.

6 DQN on the refined three-ball benchmark

I next tested DQN on the same refined physically admissible three-ball benchmark used for value iteration and tabular Q-learning.

Across the first three seeds, the best detected cycle lengths were

$$47, \quad 24, \quad 23,$$

with corresponding average rewards

$$0.777902, \quad 0.781296, \quad 0.797059.$$

Thus, two of the three DQN runs landed very close to the VI/tabular cycle-length scale, while the remaining run produced a longer cycle of length 47. The best DQN average reward matched the VI benchmark value to the recorded precision, but did not exceed it.

The length-47 DQN cycle appears to lie in the same broad stroke family as the shorter DQN, VI, and tabular-Q cycles, but in a longer and less sharply resolved realization.

7 PPO on the refined three-ball benchmark

I also tested PPO on the same refined physically admissible three-ball benchmark used for value iteration, tabular Q-learning, and DQN.

In the first small PPO sweep, all recorded best cycles had length 22. Their average rewards were below the VI benchmark value 0.797059. Thus, on the runs recorded so far, PPO recovered the same broad stroke family in cycle length and geometry, but did not match or exceed the VI average reward.

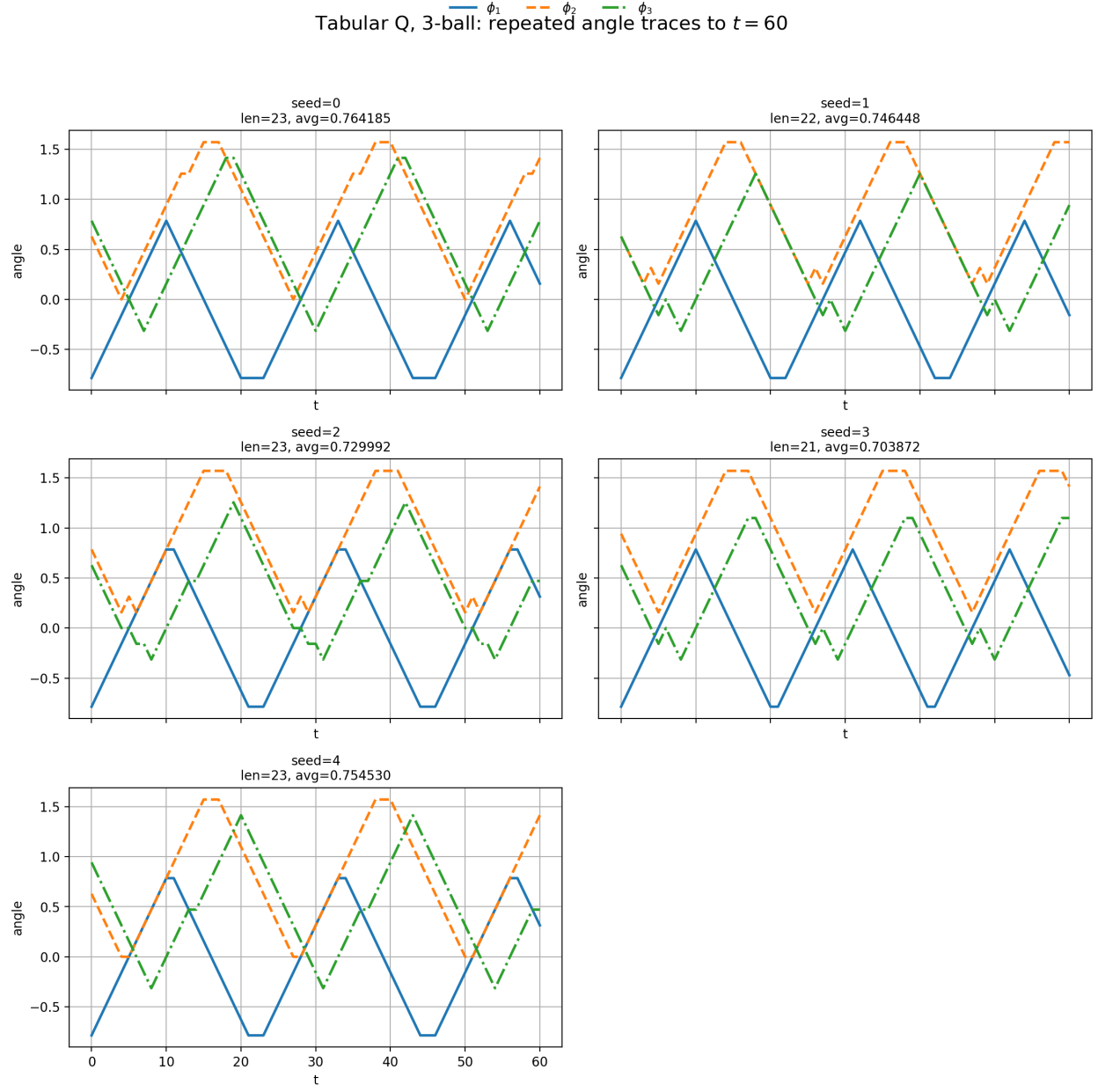


Figure 4: Repeated angle traces for the best tabular-Q cycle found at each seed.

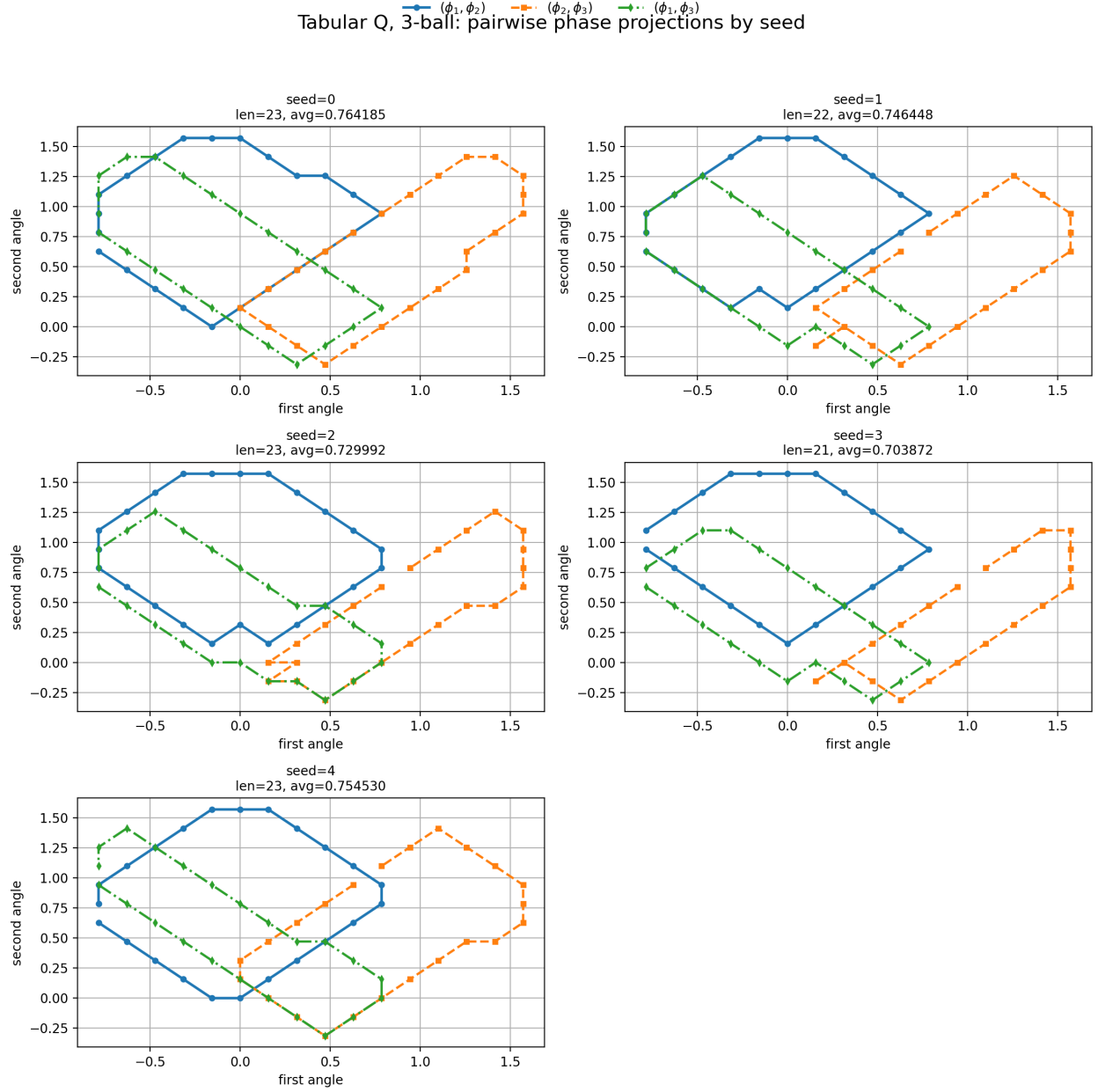


Figure 5: Pairwise phase projections for the best tabular-Q cycle at each seed.

Tabular Q, 3-ball: physical stroke overlays by seed

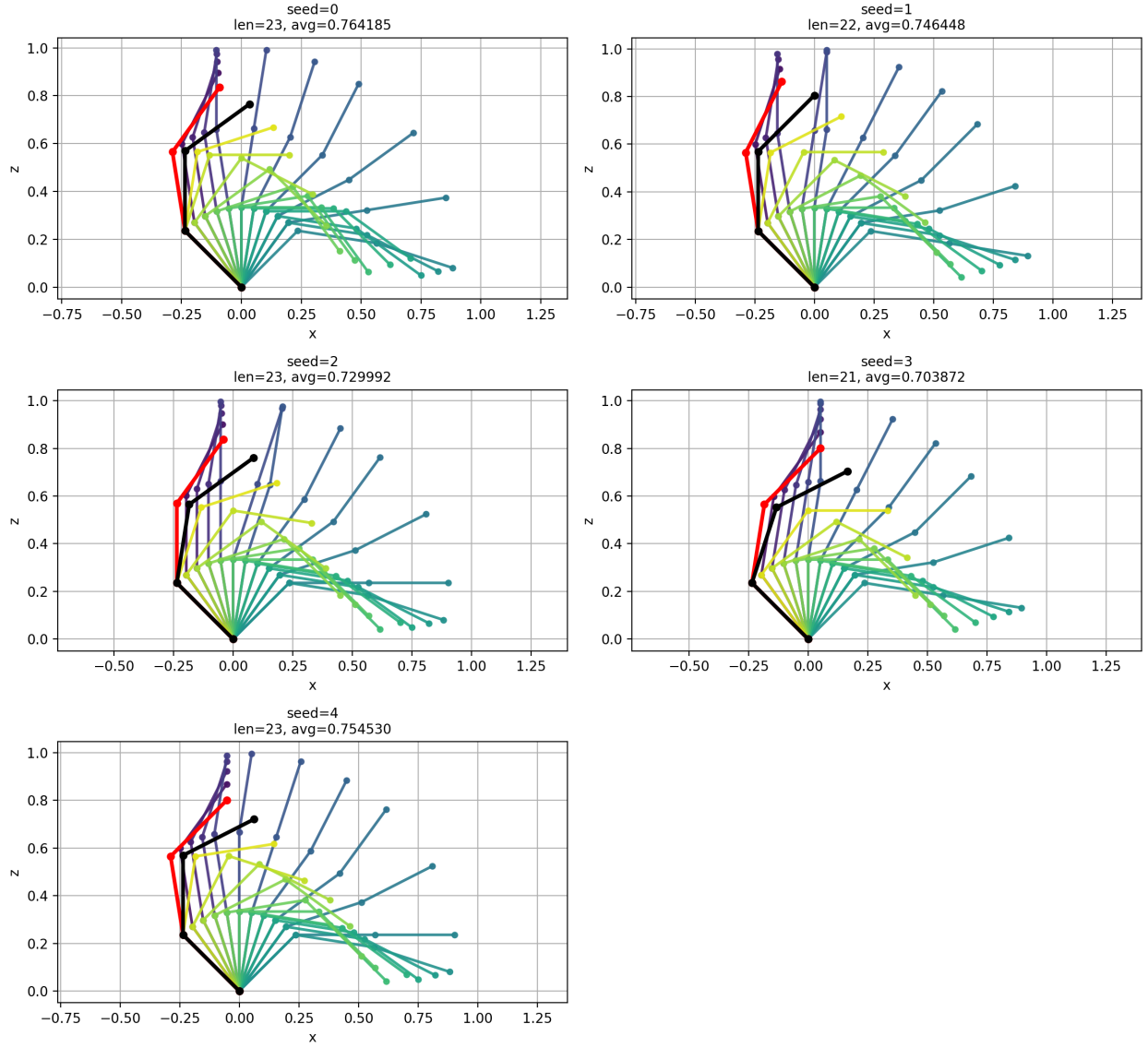


Figure 6: Physical stroke overlays for the best tabular-Q cycle at each seed.

Tabular Q, 3-ball: repeated flux reward traces to $t = 60$

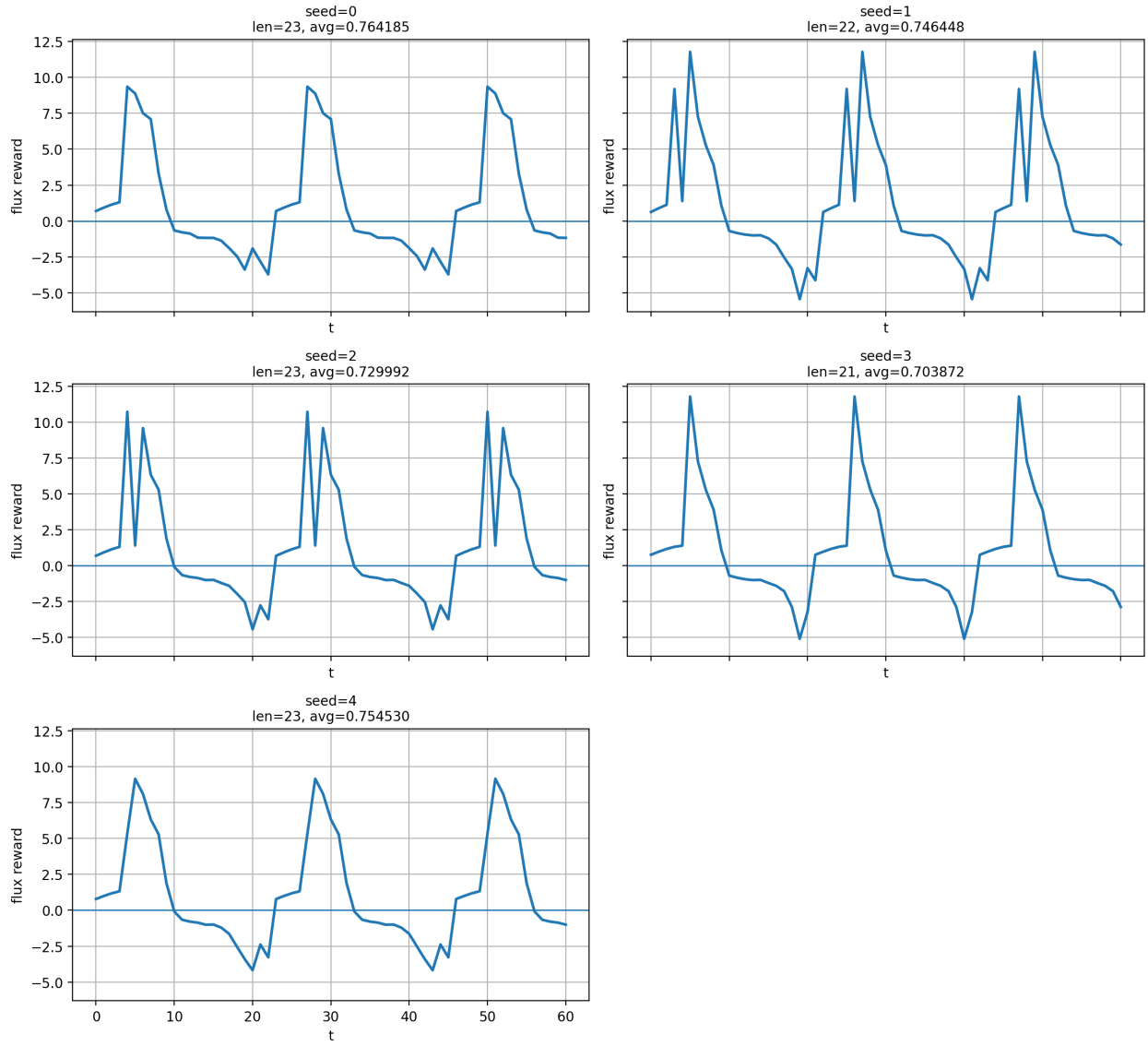


Figure 7: Repeated flux reward traces for the best tabular-Q cycles.

Tabular Q, 3-ball: best-cycle history by seed

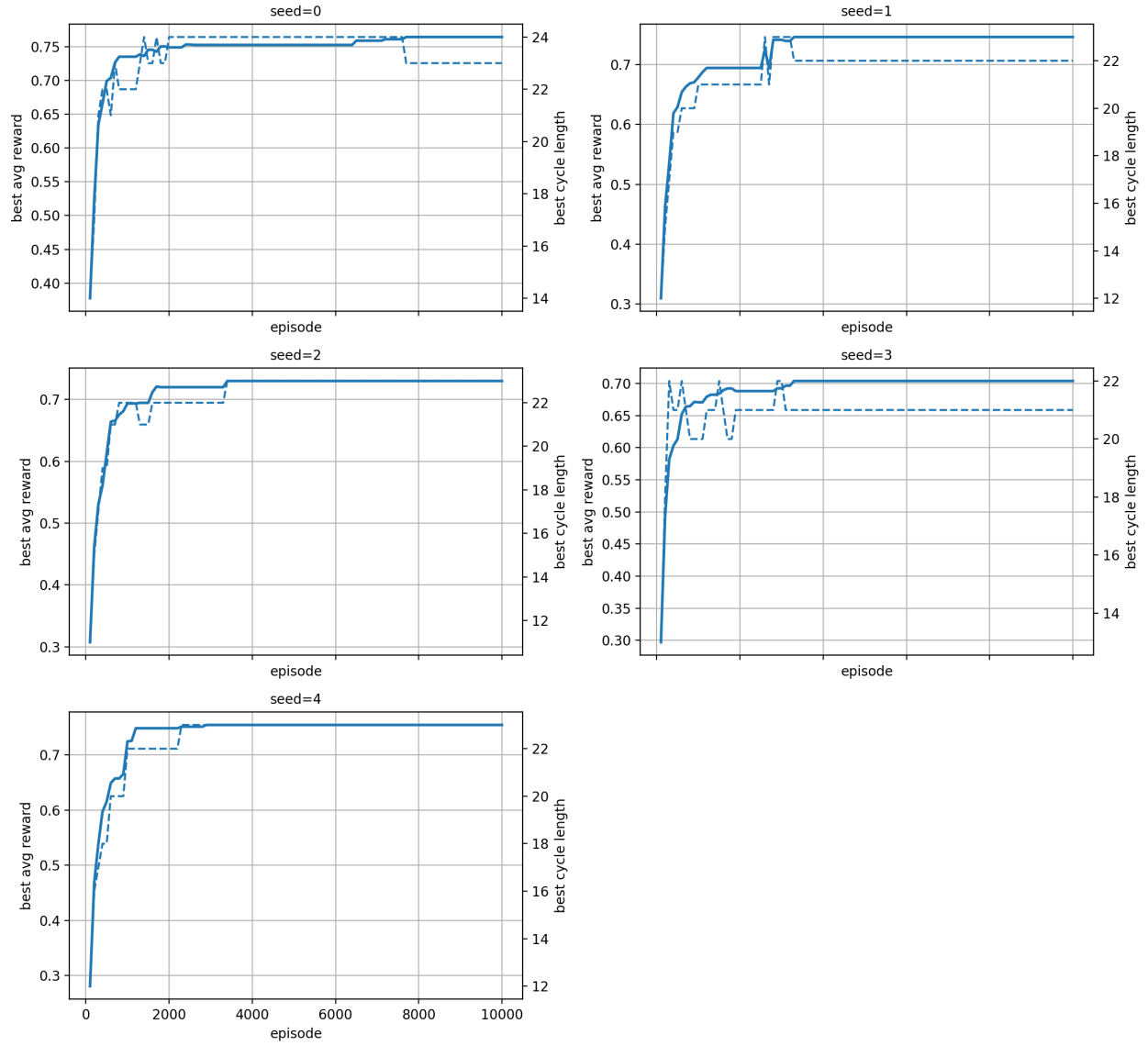


Figure 8: Evolution of the best detected greedy-policy cycle during tabular Q-learning across seeds.

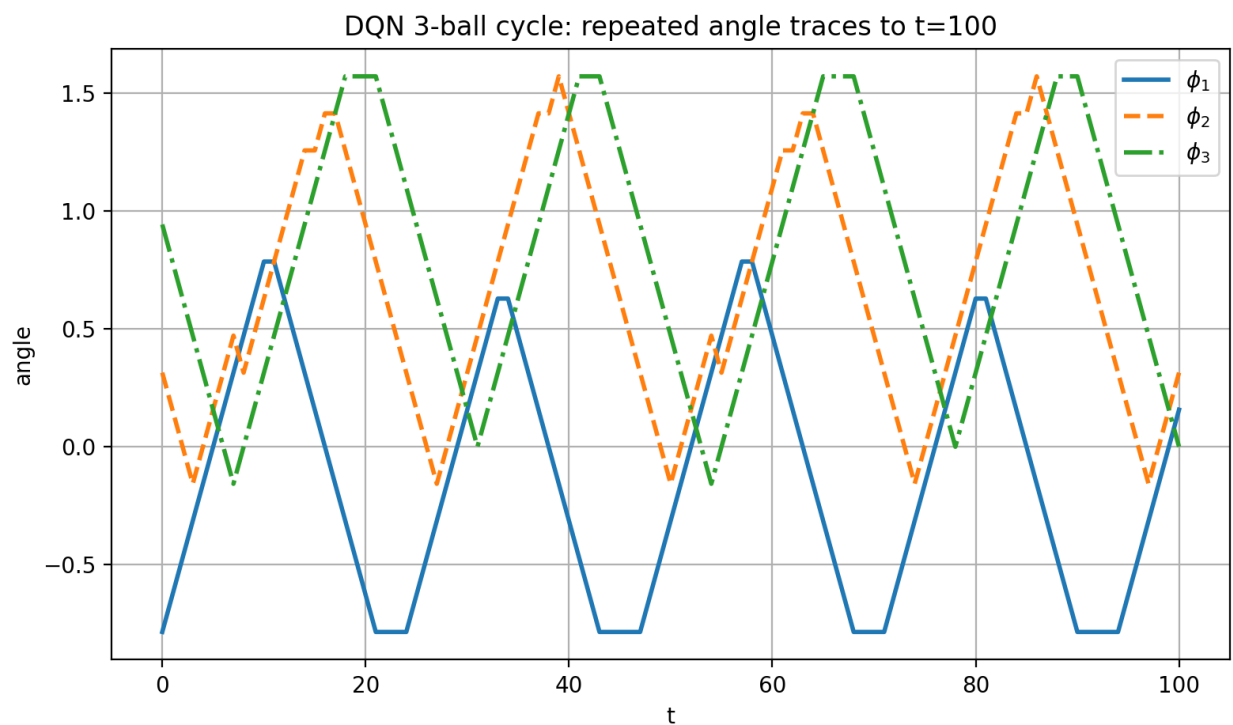


Figure 9: Repeated angle traces for the length-47 DQN cycle from seed 0.

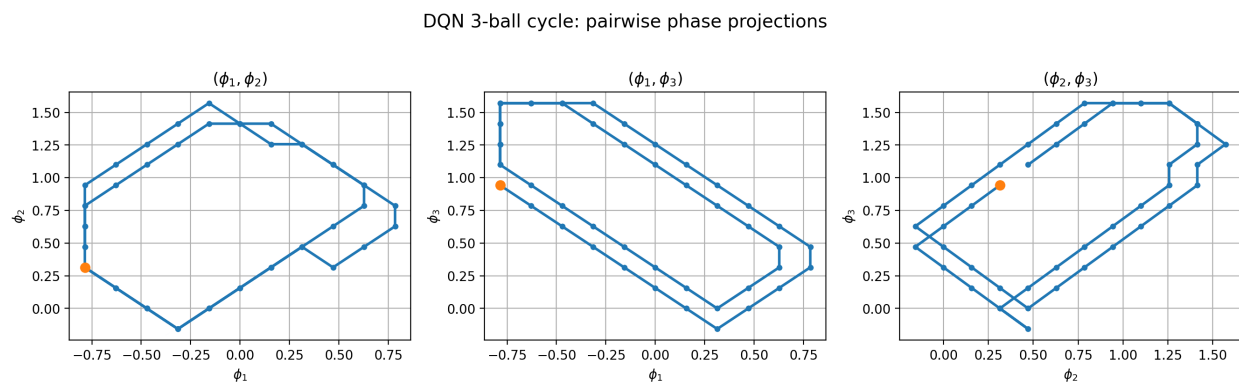


Figure 10: Pairwise phase projections for the length-47 DQN cycle from seed 0.

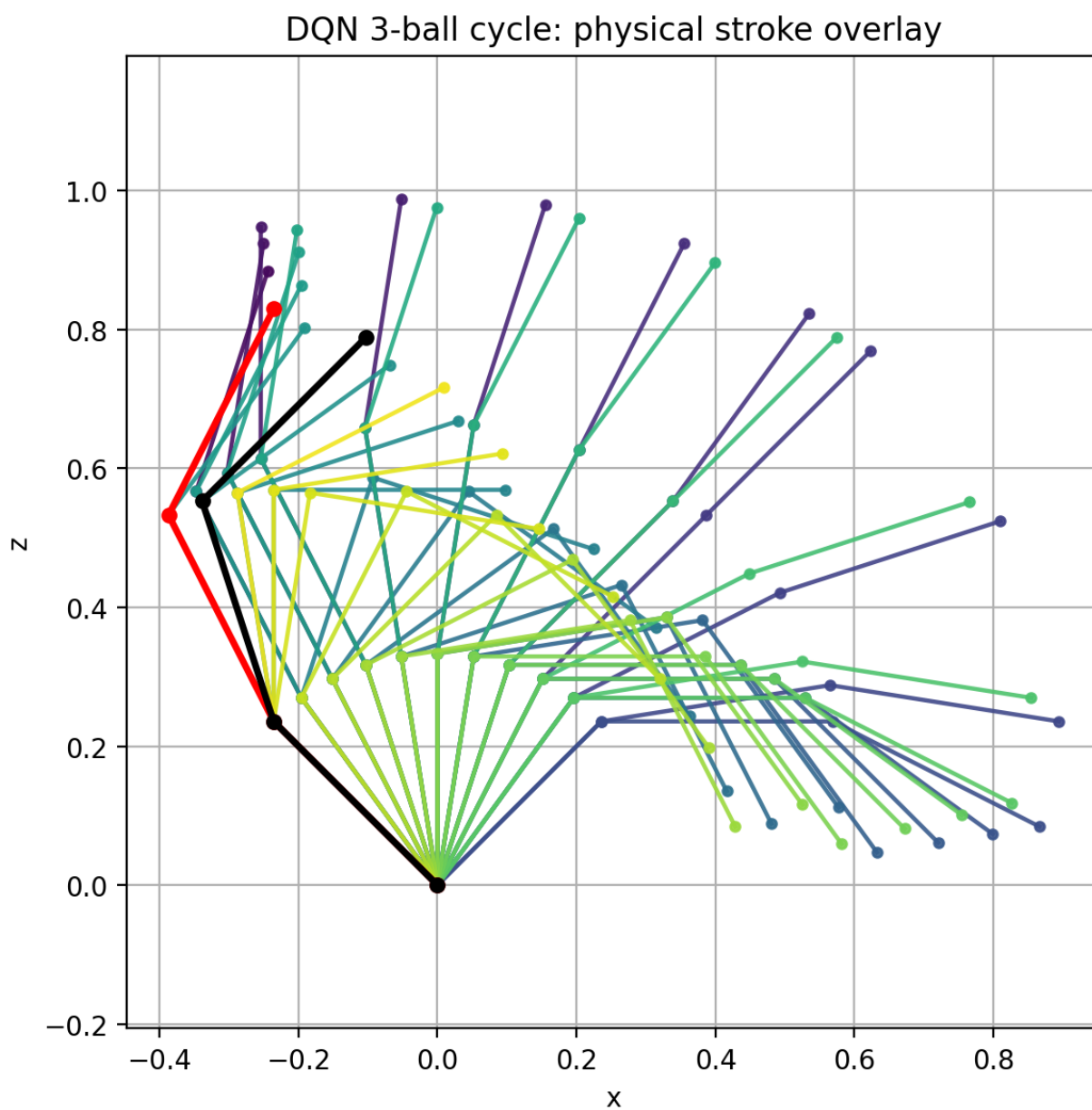


Figure 11: Physical stroke overlay for the length-47 DQN cycle from seed 0.

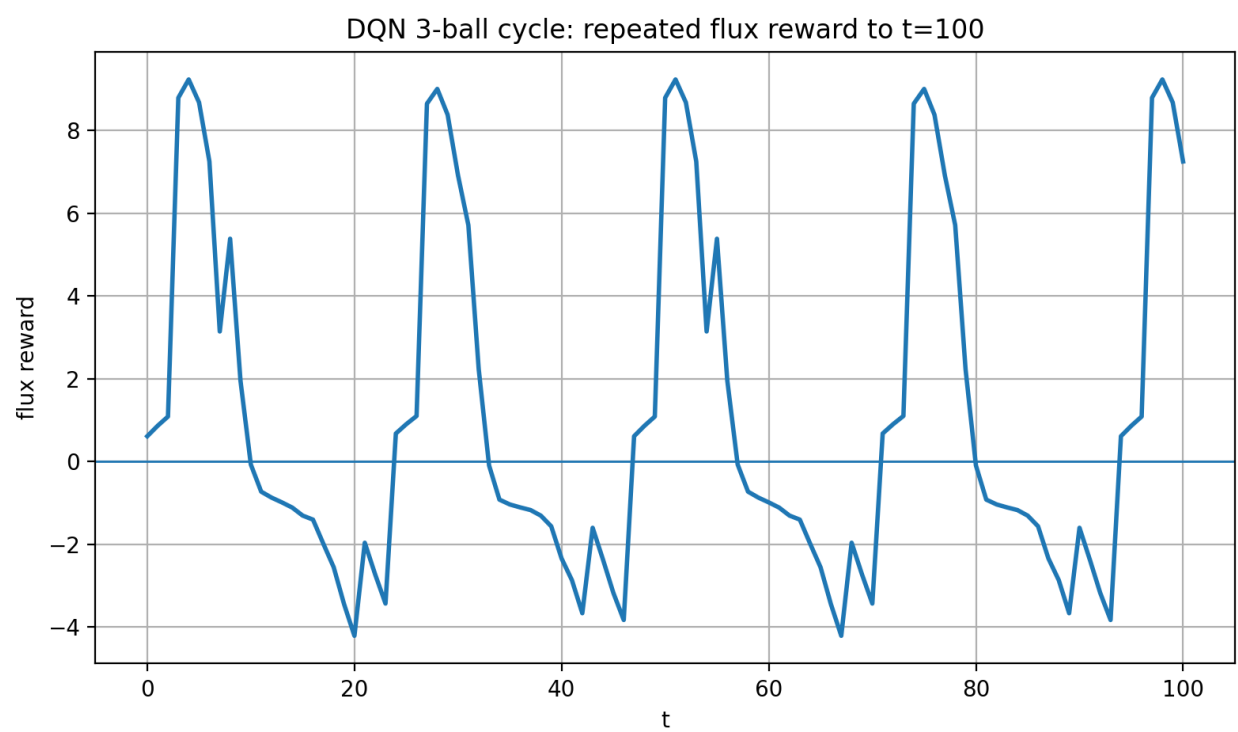


Figure 12: Repeated flux-reward trace for the length-47 DQN cycle from seed 0.

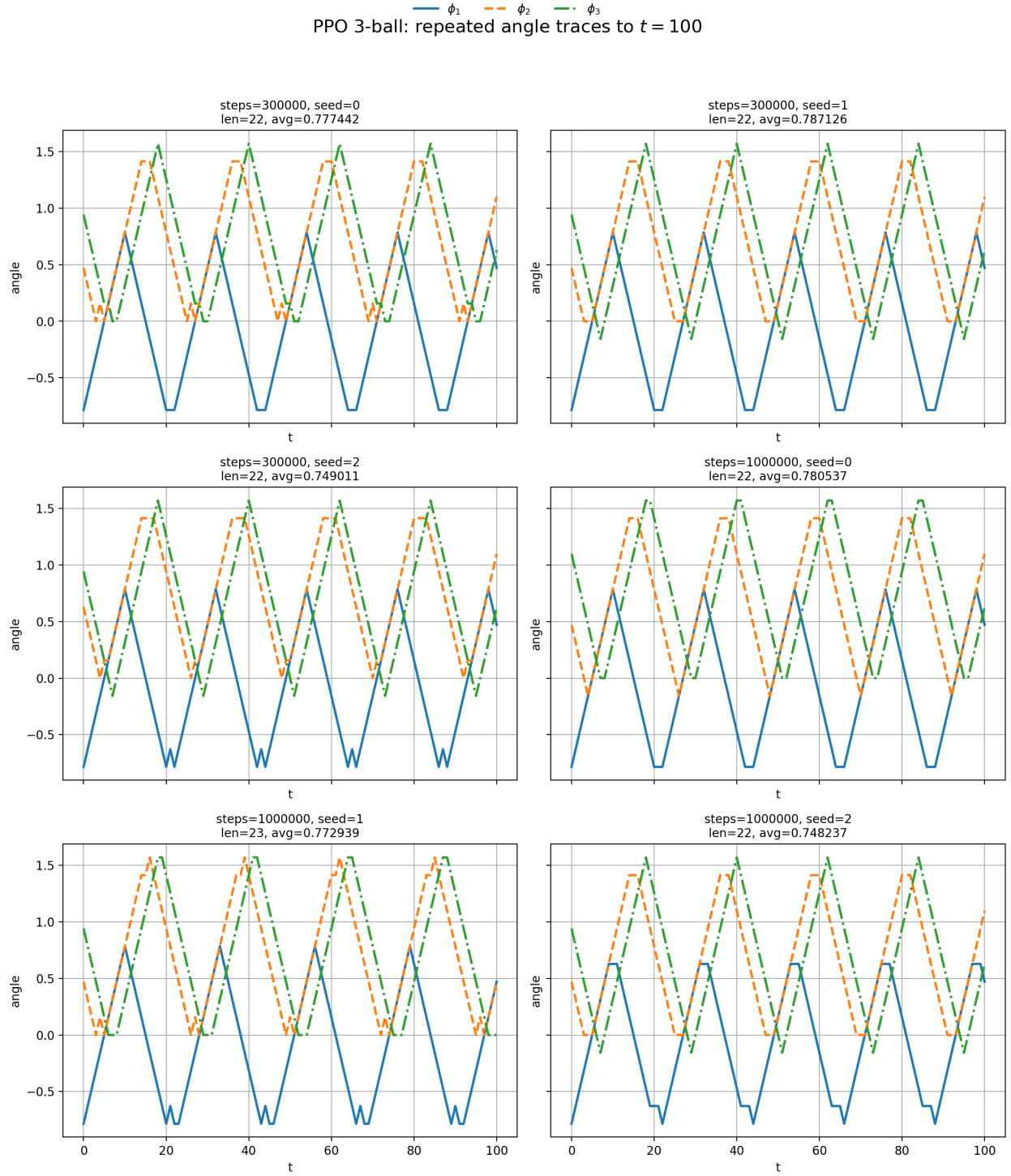


Figure 13: Repeated angle traces for the best PPO cycle in each run. The plot titles report the training timesteps, seed, cycle length, and average reward.

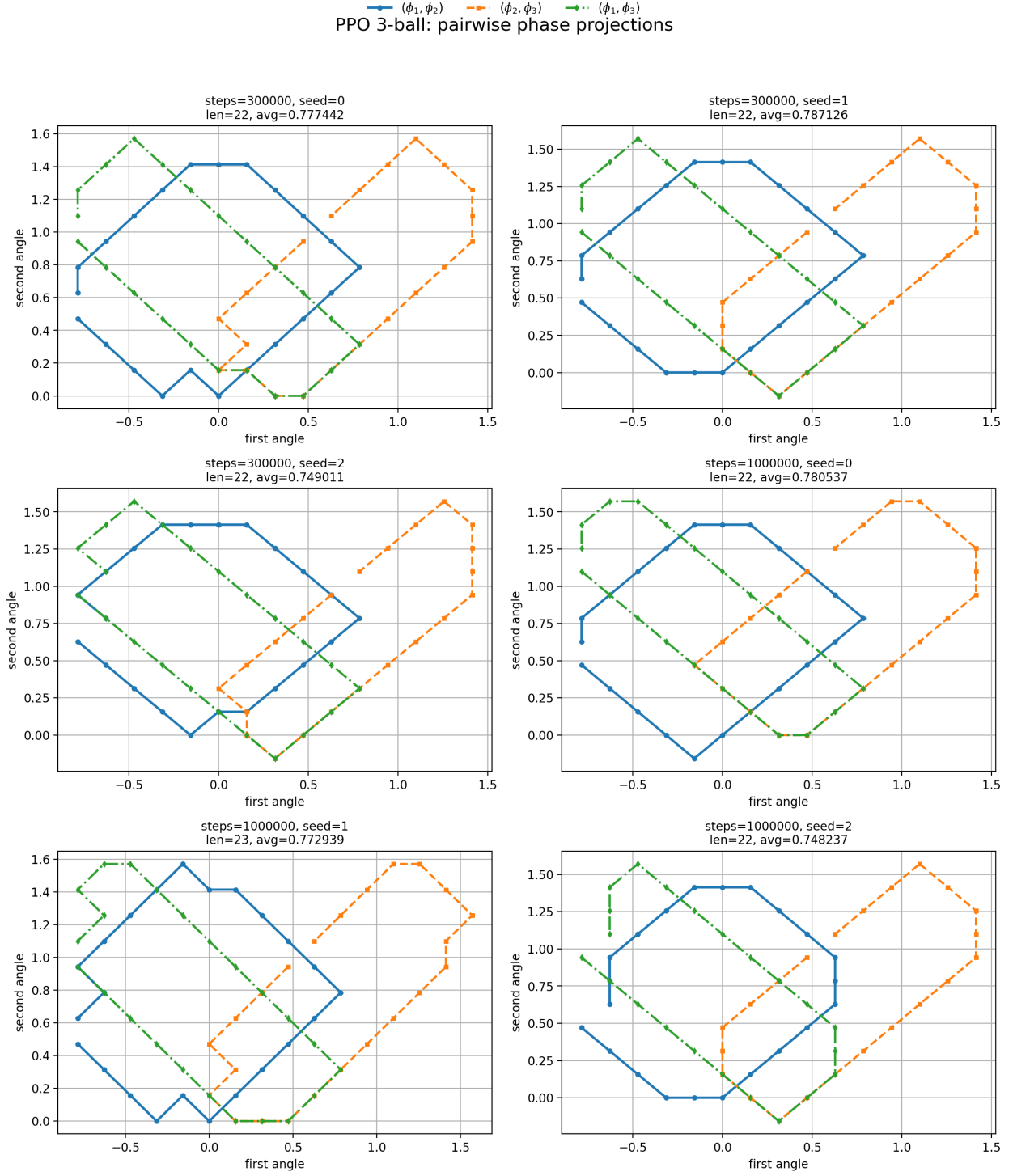


Figure 14: Pairwise phase projections for the best PPO cycle in each run.

PPO 3-ball: physical stroke overlays

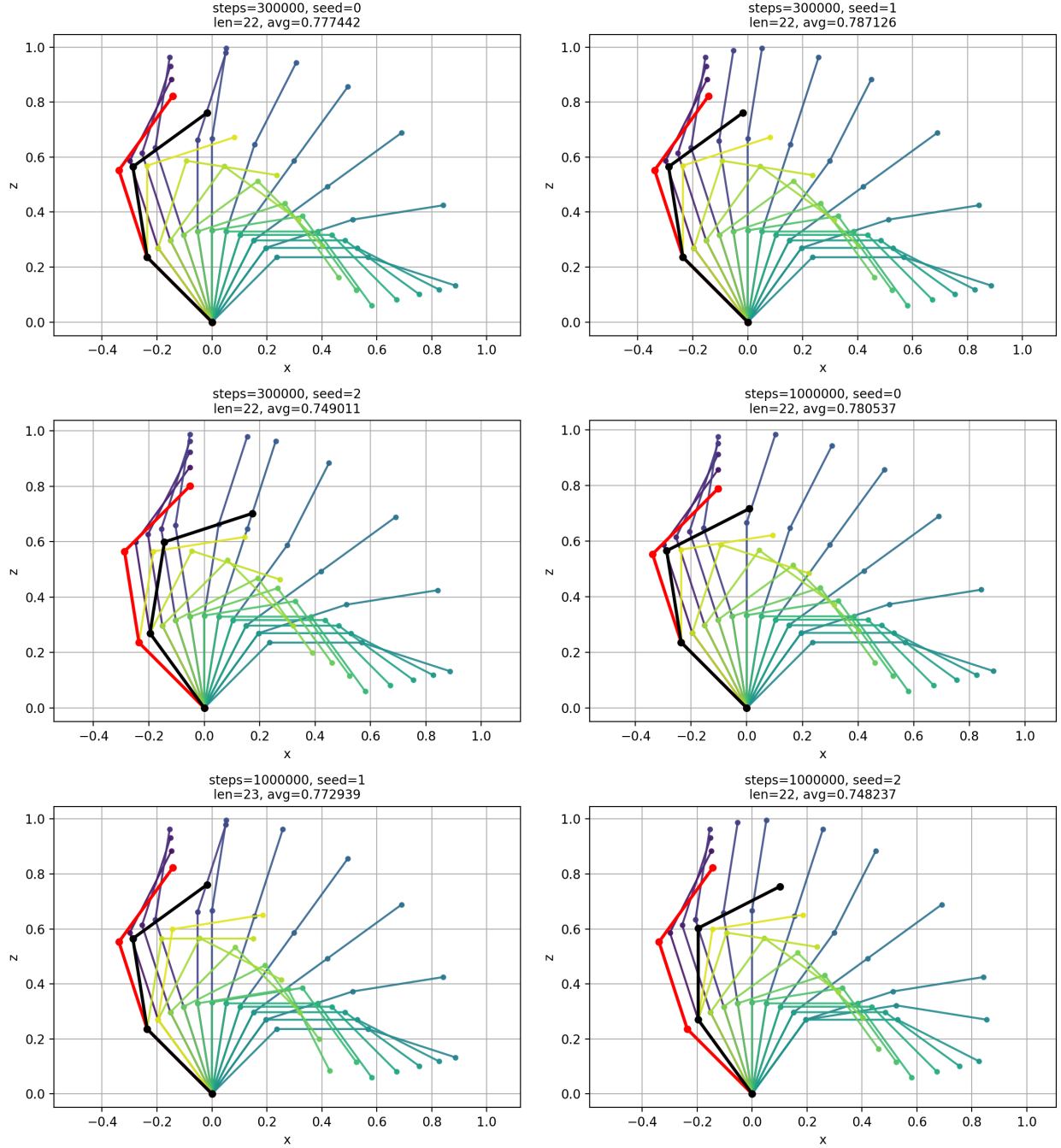


Figure 15: Physical stroke overlays for the best PPO cycle in each run.

PPO 3-ball: repeated flux reward traces to $t = 100$

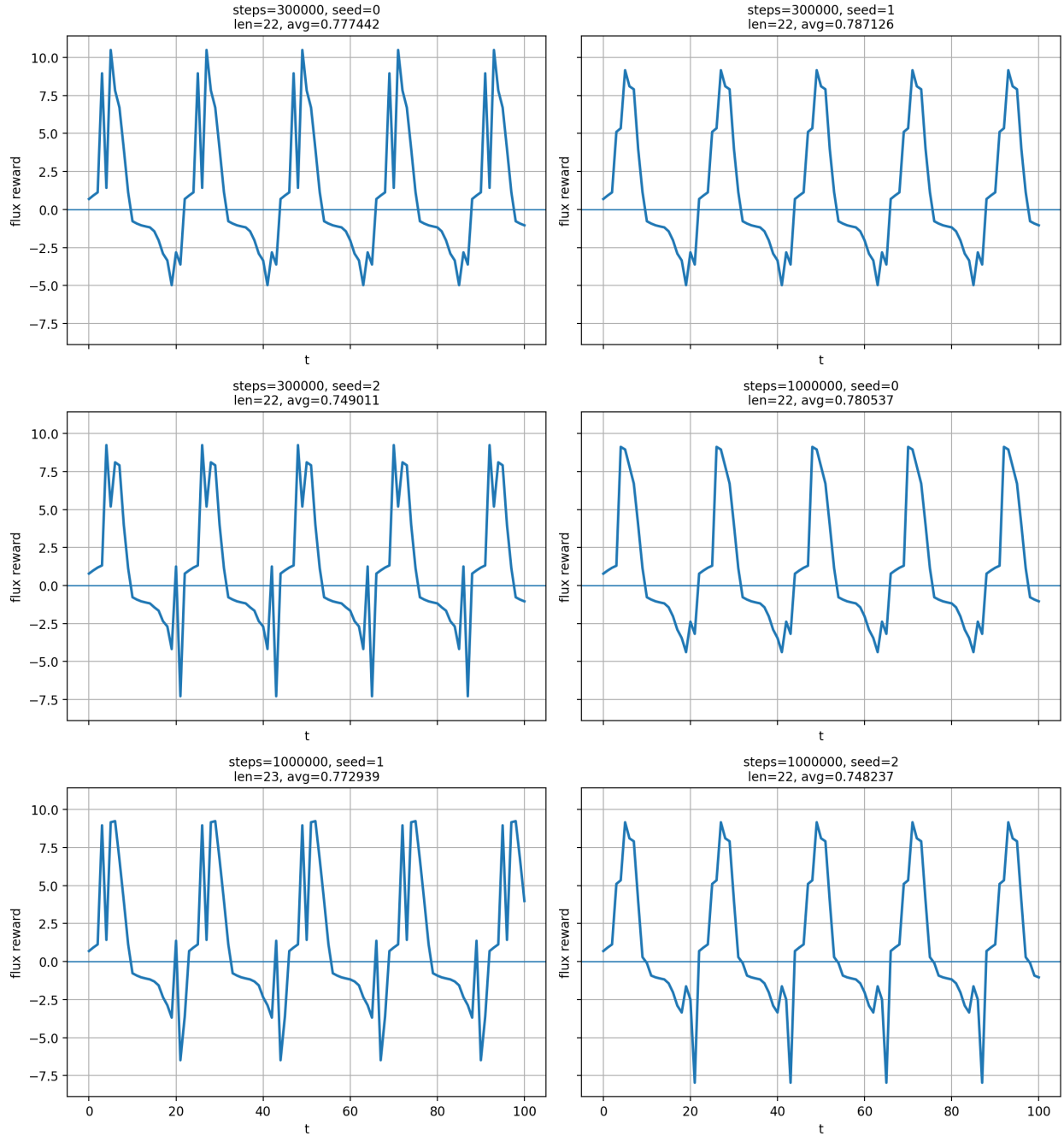


Figure 16: Repeated flux-reward traces for the best PPO cycle in each run. The plot titles report the training timesteps, seed, cycle length, and average reward.